

Design FMEA (Qualitative) — VBF-T5R3-DFMEA-01

****Case**:** t5_r3 · ****Design**:** Y4 (primary) · ****Caveat**:** qualitative version based on simulation risk signals only; severity/occurrence rated high/medium/low from the magnitude of the simulated value vs thresholds. Complete FMEA (process/supplier failures) is N/A (beyond pure simulation boundary).

#	Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	Design-side mitigation
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1	Lithium plating on negative electrode during 4C fast charge	Salt-concentration transport limit at high rate (root-caused in round 2: baseline plated severely, min_ap -0.438 V)	r6_y4_4c_df n.json:anode_potential_v` min` +0.0200 V` — positive, but margin is the thinnest of the three criteria	High	Low (no plating point in simulated CC window; margin thin at cut-off)	Transport upgrade (σ 5.0 S/m, t_{eff} 0.6, D_{eff} 9e-10), 3 μm anode particles, mild warmth ($h=26$ W/m ² K) engineered in; CV tail untested — physical validation recommended
2	Thermal overrun / temperature overshoot during fast charge	Heat generation at 4C exceeds cooling capacity	T_max_K` +327.60 K vs 333.15 K` — 5.6 K margin	High	Low	Double-sided cooling 0.01062 m ² + h tuning; margin relies on cooling-area assumption — validate in physical cell
3	Electrolyte oxidative decomposition	High cathode potential vs electrolyte HOMO	Not computed — true DFT endorsement skipped (real_compute=false`, endorsed honestly)	Medium	Unknown (honest)	Baseline EC/LiPF6-class chemistry at standard 4.2 V window (NMC811); no claim made without first-principles signal
4	Insufficient capacity / energy shortfall	Coating/geometry inadequacy	1C capacity_ah` 5.0442 vs nominal 5.0; ED 535.72 vs 500.94	Low	Low	DFN-verified; ED margin 7%
5	Manufacturing-induced performance drift	5–6 μm metal foils handling; $\pm$ tolerance effect on warmth-dependent plating margin (~6.6 mV/K sensitivity measured in round 5)	No direct signal (thermal-kinetic coupling measured: rounds 4–5 h-series)	Medium	Medium	Y3 (Al 8 μm / sep 9 μm) provided as supply-friendly alternate; DVPR flags foil handling
6	CV-tail plating at high SOC	Current taper regime not simulated in 4C protocol	Indirect: anode potential at 4.2 V cut-off (+0.0200 V, worst point of CC window); risk monotonic with current (domain reasoning)	High	Low	Physical 4C + CV charge test with reference electrode listed in DVPR

Conclusion

Highest-risk items: plating margin at CC cut-off (#1) and the cooling-assumption dependence of the thermal margin (#2). Both are design-engineered (transport + thermal + kinetic margins at DFN precision) but the *margins* — not direction — remain simulation-grade; physical validation items are named in the DVPR. Full process/supplier FMEA: **N/A (beyond pure simulation boundary)**.